

MINUTES OF MEETING (MOM)

Team 8

3D Printing Cost Estimation Platform

Meeting Details

Project: 3D Printing Cost Estimation Platform
Week: Week 11 (Final Week)
Date: Friday, April 17, 2026
Time: 4:00 PM IST
Location: CIE, Ground Floor
Client: Anil Kumar Upadhyay, Five Fingers Innovative Solutions
Client Email: aku1974india@gmail.com
Client Phone: 9949497560

Team Members Present

- Abhiroop Pullepu
- Mehrish Khan
- Ronith Menneni
- Sai Hasini
- Ruschita Guddeti

Notes

- Final review meeting for R2 (Release 2) - Production Ready.
- Client expressed satisfaction with the complete system and all deliverables.

1 R2 Final Release Status

1.1 Release Information

- **Version:** R2 (Final Release)
- **Status:** Production Ready
- **Test Coverage:** 30 test cases covering all use cases
- **Release Date:** April 17, 2026

1.2 Delivered Features

- **Core Functionality**

- STL/STEP file upload with integrity checks and file size validation (max 50MB)
- 3D Model Visualization using Three.js for STL rendering
- Cura Slicing Integration with full parameter control and G-code generation
- G-code Preview with layer-by-layer visualization
- Comprehensive cost estimation with overhead factors
- Job History with status tracking and G-code download links

- **User Management**

- Secure username/password authentication with JWT tokens
- Email verification via Mailtrap integration
- Role-based access (Customer and Admin roles)
- Profile management

- **Admin Panel**

- Material management (PLA, ABS, PETG, etc.)
- Configuration management (machine time rates, waste factors, failure factors)
- Failure logs dashboard with filtering
- Audit trail for configuration changes

- **Advanced Features**

- Model validation for printer volume constraints ($200 \times 200 \times 200$ mm)
- Slicing parameters configuration (layer height, infill, supports)
- Enhanced error handling with specific error messages
- Performance monitoring and optimization

2 Test Suite Results

2.1 Final Test Results

- **Total Tests:** 35 (30 test plan + 5 existing)
- **Passed:** 35
- **Failed:** 0
- **Skipped:** 0
- **Test Coverage:** All use cases (UC-01 through UC-15) plus integration, performance, and security tests

2.2 Test Categories Delivered

- Authentication tests (3)
- Upload tests (3)
- 3D Viewer tests (2)
- Slicing Parameters tests (2)
- Slicing Process tests (3)
- Cost Formula tests (2)
- Cost Display test (1)
- Error Handling test (1)
- Admin Logs test (1)
- Material Config tests (3)
- Machine Rate test (1)
- Overhead Factors test (1)
- Validation test (1)
- Job History test (1)
- Integration test (1)
- Performance tests (3)
- Security test (1)

3 Documentation Delivered

3.1 Technical Documentation

- Updated README.md with R2 release information
- Comprehensive feature documentation
- Cost calculation formula with examples
- Detailed project structure
- Test suite documentation with coverage breakdown
- Environment variables documentation
- Troubleshooting guide
- API documentation (accessible at /docs endpoint)

3.2 Test Documentation

- TestPlan.xls with detailed test cases
- TestPlan.csv for easy reference
- TEST_RUNNER_README.md with test execution instructions Individual test files with clear documentation

3.3 Handover Documentation

- Deployment guide
- User manual
- System architecture overview
- Installation and setup instructions
- Troubleshooting procedures

4 Client Feedback and Satisfaction

4.1 Client Review

- Client reviewed the complete system including all features, tests, and documentation.
- Client performed hands-on testing of the platform.
- Client reviewed the comprehensive test suite results.
- Client examined all documentation including deployment guides and user manuals.

4.2 Client Satisfaction

- **Feature Satisfaction:** Client is satisfied with all implemented features and their functionality.
- **Test Coverage:** Client appreciates the comprehensive test suite with 30 test cases covering all use cases.
- **Documentation Quality:** Client is satisfied with the quality and completeness of all documentation.
- **System Performance:** Client is satisfied with the performance and reliability of the system.
- **User Experience:** Client is satisfied with the user interface and overall user experience.
- **Admin Panel:** Client is satisfied with the admin panel functionality and configuration options.

4.3 Client Comments

- “The system meets all our requirements and expectations.”
- “The comprehensive test suite gives us confidence in the system’s reliability.”
- “The documentation is excellent and will help our internal team with deployment and maintenance.”
- “The admin panel provides all the configuration options we need.”
- “The cost estimation formula is accurate and transparent.”
- “We are satisfied with the final deliverables and ready to proceed with deployment.”

5 Project Closure

5.1 Deliverables Checklist

Complete 3D Printing Cost Estimation Platform (R2)

30 test cases with 100% pass rate

Comprehensive documentation

Deployment guide

User manual

Test plan documentation

Source code on GitHub

Docker Compose configuration

5.2 Known Limitations (Documented)

- STEP file support is planned but not yet implemented (STL only)
- Account lockout after failed login attempts is not implemented
- G-code preview animation is basic (layer-by-layer only)
- 3D viewer supports STL files only (STEP files require conversion)

5.3 Future Enhancements (Optional)

- STEP file support with STEP → STL conversion
- Multiple printer profiles with different volumes and rates
- Enhanced G-code preview with animation
- Account lockout after failed login attempts

6 Final Action Items

Owner	Task
Team 8	Provide final source code access and documentation to client IT team.
Team 8	Schedule deployment support session with client IT team.
Client (Anil Kumar)	Review and approve final deliverables.
Client (Anil Kumar)	Proceed with production deployment.

7 Project Conclusion

PROJECT COMPLETED SUCCESSFULLY - R2 FINAL RELEASE

- All requirements met and delivered
- Client is satisfied with all deliverables
- System is production-ready
- Documentation is complete
- Test suite is comprehensive with 100% pass rate

Minutes prepared by: Team 8
Date: April 17, 2026